

# Product Laws of Exponents

*Teacher's Edition*

## Read This First

This lesson rebuilds Math Nation 3.1 (SE pp. 99–105) as front-taught notes and individual boards. The book's version is four partner Explorations in a row; every table in them survives here, worked from the front, because a single student can complete each one.

The mathematics in the book's lesson is clean — every value on its pages checks. The defect is in its practice key: the Additional Practice answer for  $(-3 \cdot 6)^4$  is printed 104,796 and is 104,976. That item is on the Additional bank here with the right value.

Two things the state guide expects that the book never asks: working backwards to an unknown exponent ( $2^3 \cdot 2^n = 2^8$ ), and the sum-inside-the-parentheses trap  $((10 + (-2))^3)$ . Both are in the round and in the bank.

## Standard and Targets

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

**Clarification** Instruction builds the laws from specific examples; Appendix E lists them. Whole-number exponents only in this lesson — the zero exponent arrives in 3.02 and negative exponents in 3.04.

**Boundary** Rational-number bases, including negative and decimal bases. No variables until the Thread A days after 3.07.

**Learning target** I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

**Essential question** How can the laws of exponents be used to write an equivalent expression?

**Building on** MA.6.NSO.3.3 — evaluating positive rational numbers with natural-number exponents (Grade 6). Unit 2 — squares and cubes as repeated multiplication.

**Working toward** MA.8.NSO.1.3 (integer exponents, Lessons 3.04–3.07) and MA.8.AR.1.1 (the same laws with variable bases, the Thread A days after 3.07).

**Where this sits in the course** First of three lessons carried from MA.7.NSO.1.1 (3.01–3.03). Everything here is rehearsed again with integer exponents under MA.8.NSO.1.3, which the Grade 8 FAST reports under Number Sense and Operations.

## Lesson at a Glance

---

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **16 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2-3    | 5 min         |
| Notes                               | 4-7    | 14 min        |
| Example 1                           | 8-9    | 3 min         |
| Your Turn                           | 10-11  | 3 min         |
| Example 2                           | 12-13  | 3 min         |
| Your Turn                           | 14-15  | 3 min         |
| Whiteboards                         | 16-33  | 16 min        |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

**Materials** Whiteboards and markers. Calculators are allowed on everything and are useful for the big values in the bank ( $8^8$ ,  $35^4$ ); the laws themselves are done by hand.

**Where the lesson lives** Notes II (the expansion that proves the product law) and Whiteboard questions 5 and 6 (the two diagnostics that separate 'add' from 'multiply'). If the round shows a board full of B on question 5, the expansion is what to go back to.

**No partner or group work anywhere in this lesson.** Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

## Vocabulary

---

**base** — the number that is used as a factor: in  $2^3$  the base is 2

**exponent** — the number of times the base is used as a factor:  $2^3 = 2 \cdot 2 \cdot 2$

**power** — a base with an exponent, such as  $2^3$ ; its value is 8

**product of powers law** —  $a^m \cdot a^n = a^{m+n}$  — same base, add the exponents

**power of a product law** —  $(ab)^m = a^m \cdot b^m$  — the exponent goes to every factor

**power of a power law** —  $(a^m)^n = a^{mn}$  — multiply the exponents

**identity exponent law** —  $a^1 = a$

## Warm-Up — spaced retrieval

---

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1.  $\sqrt{81}$      9

retrieves: yesterday — Unit 2 — the principal square root (2.02)

2. Between which two whole numbers is  $\sqrt{40}$  ? **6 and 7**

retrieves: last week — Unit 2 — approximating roots (2.04):  $36 < 40 < 49$

3. Solve  $3x - 7 = 11$   **$x = 6$**

retrieves: last unit — Unit 1 — two-step equations (1.04)

4. Evaluate  $\left(\frac{2}{3}\right)^2$   **$\frac{4}{9}$**

retrieves: prior grade — the prerequisite this lesson needs — MA.6.NSO.3.3 — a fraction raised to a whole-number power

## Slide-by-Slide Teaching Notes

---

### Slide 1. Product Laws of Exponents 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

### Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the prerequisite: a rational base is squared by squaring numerator and denominator. A board showing  $4/3$  or  $2/9$  tells you the base was not treated as one number.

### Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

### Slide 4. Notes — Copy the three words. 3 min

- Students copy the three vocabulary rows. Say them aloud: base, exponent, power. The exponent COUNTS factors; it is not a factor itself — that is the misconception behind  $2^3 = 6$ .
- **OFF THE SLIDE, YOURS TO SAY:** Ask: what does 2 to the 3 mean? Insist on the answer 'three 2s multiplied', not 'six'.

### Slide 5. Notes — Where the rule comes from. 4 min

- Expand both powers, count the factors, then write the rule. The law only applies when the bases match:  $3^2 \cdot 2^3$  has no shortcut, and the reveal on the second row is the reason.
- **OFF THE SLIDE, YOURS TO SAY:** 'We are not adding the bases and not multiplying the exponents. We are counting factors.'

### Slide 6. Notes 3 min

- Expand  $(2 \cdot 5)^3$  as three copies of  $(2 \cdot 5)$ , regroup with the commutative and associative properties, and read off  $2^3 \cdot 5^3$ . The trap is a SUM inside the parentheses:  $(10 + (-2))^3$  is  $8^3$ , and the exponent does not distribute over addition. Question 9 of the round tests exactly this.

### Slide 7. Notes 4 min

- $(4^2)^3$  is three copies of  $4^2$ , so six 4s: multiply the exponents. Then the identity law: one factor of the base is the base. Students who write  $4^2 \cdot 3 = 12$  or  $4^5$  (adding) are confusing this with the product law; the reveal on the table makes the count visible.

- **OFF THE SLIDE, YOURS TO SAY:** 'Product of powers: add. Power of a power: multiply. Both come from counting factors, so count if you forget.'

**Slide 8. Example 1 — Two ways to the same value.** 1 min

- SPOKEN ONLY. Read the expression, ask which law applies and why (same base). Do not write yet.

**Slide 9. Example 1 — Worked both ways.** 2 min

- Left column: the law, one exponent, then the value. Right column: evaluate each power and multiply. Same 243 either way — that is the check students can always run.

**Slide 10. Your Turn — Same steps, your numbers. Boards up when done.** 2 min

- Catches a student who multiplied the exponents ( $2^{12} = 4096$ ) or multiplied the bases ( $4^7$ ). A board reading 128 or  $2^7$  is right.

**Slide 11. Your Turn — Answer.** 1 min

- Reveal; name what a wrong board most likely did (see the note above).

**Slide 12. Example 2 — A negative base and a power of a power.** 1 min

- SPOKEN ONLY. The brackets are not absolute value bars — say so before anyone asks. Ask: how many factors of  $(-2)$  are there in total?

**Slide 13. Example 2 — Worked.** 2 min

- Multiply the exponents: six factors of  $-2$ . An even number of negative factors gives a positive product; the sign question is the one students carry into Lesson 3.02.

**Slide 14. Your Turn — Same steps, your numbers. Boards up when done.** 2 min

- Catches the added-exponent error ( $(-3)^5 = -243$ , a negative answer) and the dropped sign (a student who writes  $3^6$  gets 729 too — ask them to write the base with its sign).

**Slide 15. Your Turn — Answer.** 1 min

- Reveal; name what a wrong board most likely did (see the note above).

**Slide 16. Whiteboards · Question 1 of 9** 16 min for the round

- Product of powers, the plain case. A board with  $25^5$  multiplied the bases;  $5^6$  multiplied the exponents.

**Slide 17. Whiteboards · Question 1 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 18. Whiteboards · Question 2 of 9**

- Same law, now evaluated.  $2^8 = 256$  on a board means the exponents were multiplied;  $4^6$  means the bases were.

**Slide 19. Whiteboards · Question 2 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 20. Whiteboards · Question 3 of 9**

- Power of a product. Either route lands on 1000. A board with 30 multiplied 10 by 3.

**Slide 21. Whiteboards · Question 3 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 22. Whiteboards · Question 4 of 9**

- Power of a power.  $3^6$  means the exponents were added.

**Slide 23. Whiteboards · Question 4 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 24. Whiteboards · Question 5 of 9**

- Diagnostic. B is the most common wrong board; a board full of B means the two laws are being told apart by the LOOK of the expression rather than by counting factors — go back to the Notes II expansion.

**Slide 25. Whiteboards · Question 5 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 26. Whiteboards · Question 6 of 9**

- Diagnostic, mirror of Q5. D is rare but instructive: the student treated the exponent as a number to be squared.

**Slide 27. Whiteboards · Question 6 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 28. Whiteboards · Question 7 of 9**

- Same base with a sign. A board with 27 dropped the sign; a board with  $-3^3$  read as  $-27$  is fine but ask for the parentheses.

**Slide 29. Whiteboards · Question 7 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 30. Whiteboards · Question 8 of 9**

- Rational base. This is the B1G-M example item for 7.NSO.1.1. A board with  $1/10$  multiplied the bases' denominators by the count;  $1/6$  multiplied exponents and used  $\frac{1}{2}$  once.

**Slide 31. Whiteboards · Question 8 of 9**

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

**Slide 32. Whiteboards · Question 9 of 9**

- Written. The law needs a product; a sum is evaluated first. Accept any answer that says 'add first' and lands on 512. A student who writes  $10^3 + (-2)^3 = 992$  has

distributed the exponent over addition — the misconception this question exists to surface.

### Slide 33. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

### Slide 34. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

## The Whiteboard Round — what each board tells you

---

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

| # | Question                        | Answer                      | What a wrong board says                                                                                                                                                                                                                                                                                                   |
|---|---------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $5^2 \cdot 5^3$                 | $5^5$                       | $25^5$ — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; $5^6$ — multiplied the exponents [confused with power of a power]                                                                                                                                                                        |
| 2 | $2^4 \cdot 2^2$                 | <b>64</b>                   | 256 — multiplied the exponents ( $2^8$ ); 4096 — multiplied the bases ( $4^6$ )                                                                                                                                                                                                                                           |
| 3 | $(2 \cdot 5)^3$                 | <b>1000</b>                 | 30 — multiplied the base by the exponent [misconception: exponent treated as a factor]; 250 — applied the exponent to one factor only                                                                                                                                                                                     |
| 4 | $(3^2)^4$                       | $3^8$                       | $3^6$ — added the exponents [confused with product of powers]; $9^8$ — squared the base as well                                                                                                                                                                                                                           |
| 5 | Which expression is equivalent? | <b>A — <math>4^5</math></b> | (B) multiplied the exponents — the power-of-a-power move applied to a product [7.NSO.1.1 misconception, B1G-M]; (C) multiplied the bases as well as adding the exponents [7.NSO.1.1 misconception: operating on the bases]; (D) added the bases and added the exponents [7.NSO.1.1 misconception: operating on the bases] |

| # | Question                                                                                        | Answer                                             | What a wrong board says                                                                                                                                                                                                                                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | Which expression is equivalent?                                                                 | <b>A — <math>2^6</math></b>                        | (B) added the exponents — the product-of-powers move applied to a power of a power [7.NSO.1.1 misconception, B1G-M]; (C) raised the base to the outer exponent as well [7.NSO.1.1 misconception: operating on the bases]; (D) raised the inner exponent to the outer one, $3^2 = 9$ [misconception: the outer exponent applied to the exponent, not to the power] |
| 7 | $(-3)^2 \cdot (-3)^1$                                                                           | <b>-27</b>                                         | 27 — the sign was dropped when the odd power was evaluated [7.NSO.1.1 clarification: rational-number bases include negatives]                                                                                                                                                                                                                                     |
| 8 | $\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$                                   | <b><math>\frac{1}{32}</math></b>                   | 1/6 — multiplied the exponents; 5/2 or 1/10 — treated the fraction as two numbers                                                                                                                                                                                                                                                                                 |
| 9 | Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$ . Is she right? | <b>No — <math>(10 + (-2))^3 = 8^3 = 512</math></b> | 992 — distributed the exponent over the sum [B1G-M 7.NSO.1.1: the laws apply to products and quotients, never to sums]                                                                                                                                                                                                                                            |

## Question Bank — variation and answers

**Variation.** Bank questions 1–5 vary one thing at a time inside the product law: two powers, then three, then evaluate-and-multiply against the law, then a negative base, then a case the student must judge (Jose, same base). Questions 6–8 move to the power of a product with the same three moves. The pattern breaks at question 13, where the exponent is the unknown, and at 15, where an equivalent expression has a different base ( $4^3 = 2^6$ ). The Additional sheet mirrors the structure with different numbers and breaks at Leeland (different bases — the law does not apply).

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

### Question Bank

- $(3 \cdot 3 \cdot 3)(3 \cdot 3 \cdot 3 \cdot 3) = 3^7$
- $5 \cdot (5 \cdot 5 \cdot 5 \cdot 5)(5 \cdot 5 \cdot 5) = 5^8$
- $4 \cdot 64 = 256$**
- $(-8)^8 = 16,777,216$
- Yes — the bases are the same, so  $9^3 \cdot 9^{12} = 9^{15}$  .**
- $(4 \cdot 5)(4 \cdot 5) = (4 \cdot 4)(5 \cdot 5) = 4^2 \cdot 5^2$
- $(-35)^4 = 1,500,625$
- $3^2 \cdot (-6)^2 = 9 \cdot 36 = 324$

9.  $3^{16}$
10.  $0.5^2 \cdot (-8)^2 = 0.25 \cdot 64 = 16$  — *power of a product*
11. (a) 285.6 (b)  $\frac{24}{8} = 3$
12. (a)  $n = 5$  (b)  $n = 5$  (c)  $n = 4$
13. A A
14. A, B, C, D A, B, C and D

### Question Bank — Additional

1.  $(4 \cdot 4)(4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4^7$
2.  $7^6$
3.  $125 \cdot 25 = 3,125$
4.  $(-6)^9 = -10,077,696$
5. *No — the bases are different (7 and 3), so the law does not apply. The value is  $343 \cdot 2,187 = 750,141$ , found by evaluating each power.*
6.  $(7 \cdot 8)(7 \cdot 8) = (7 \cdot 7)(8 \cdot 8) = 7^2 \cdot 8^2$
7.  $(-18)^4 = 104,976$
8.  $2^3 \cdot (-9)^3 = 8 \cdot (-729) = -5,832$
9.  $5^{16} = 152,587,890,625$
10.  $2^{10} = 1,024$  — *power of a power*
11. (a)  $n = 5$  (b)  $n = 4$  (c)  $n = 5$
12. A A
13. A, B, C, D A, B, C and D

## What the audit against Math Nation found

---

- Math Nation 3.1 is sound page by page; all its worked values were recomputed ( $6^{12}$ ,  $(-8)^9$ ,  $(-24)^5$ ,  $6^8$ ,  $10^{12}$ ,  $0.5^7$ , 16).
- Additional Practice L1 #7 key:  $(-18)^4$  printed 104,796; correct 104,976. Fixed on the Additional bank here.
- Additional Practice L1 #10 key stopped at  $2^{10}$  without the value the column asked for; the bank item asks for and keys 1,024.
- The book's warm-up is a growth-mindset video with no mathematics in it; replaced by spaced retrieval.
- The book's explorations are all 'with your partner'; rebuilt as front-taught notes and boards (house rule 4).

### What changed from the book

- 3.1.1 video warm-up → four retrieval questions.
- 3.1.2–3.1.5 partner explorations → Notes I–IV and two worked examples with Your Turns.
- 3.1.6 Your Turn and 3.1.7 Ticket Out the Door → the nine-question round; no exit ticket (ruling 10).



- Homework and Additional Practice → the two question banks, posted not assigned.